

Project Title: Safety-Aware and Gender-Sensitive Urban Route Planning: An AI Search Approach

1. Introduction and Background

Modern navigation systems (such as Google Maps) are highly efficient at finding the shortest or fastest routes between two points. However, these systems inherently assume that all paths are equally safe. In real-world urban environments, safety is a critical factor for pedestrians and drivers. Factors such as historical incident rates, street lighting, and neighborhood crime statistics heavily influence route choice. Furthermore, the perception and reality of risk can vary significantly based on gender; a poorly lit alleyway at night may carry a drastically different risk factor for a female traveler compared to a male traveler.

This project proposes the development of an intelligent urban routing system that transcends traditional distance-based navigation. By utilizing OpenStreetMap (OSM) data and synthetic incident datasets, this project formulates a novel AI search problem. The system will calculate an optimal path that dynamically balances road length, travel time, and a gender-sensitive safety metric using both uninformed and informed search algorithms.

2. Project Objectives

- * **System Development:** Build a map-based routing system using open-source OSM data to accurately map road networks, intersections, and every road turn.
- * **Synthetic Data Integration:** Generate and map synthetic "incident data" across the road network to simulate real-world safety hazards.
- * **Novel Problem Formulation:** Define a comprehensive Action Cost function that calculates a customized weight based on distance, time, and a dynamic Gender-Risk equation.
- * **Heuristic Design (Primary Challenge):** Formulate an admissible heuristic function, $h(n)$, that mathematically incorporates safety metrics without violating the condition of overestimation.
- * **Algorithmic Comparison:** Implement and compare Uninformed Search (e.g., Uniform Cost Search/Dijkstra) and Informed Search (e.g., A* Search) to observe algorithmic efficiency.

3. Formal AI Problem Formulation

To solve this pathfinding problem using Artificial Intelligence, the environment is defined using the standard five-component problem formulation:

- * State Space: The set of all valid intersections (nodes) and connecting roads (edges) extracted from the OpenStreetMap (OSM) data.
- * Initial State (S_{start}): The user's starting coordinate on the map.
- * Goal State (S_{goal}): The user's desired destination coordinate.
- * Actions: Moving from the current node to any valid, adjacent intersection (navigating road turns).
- * Action Cost Function ($g(n)$): This is the core novelty of the project. The cost of traversing a road segment from Node A to Node B is not just distance; it is a weighted equation of three variables:
$$\text{Cost}(A, B) = (\alpha * \text{Distance}) + (\beta * \text{Time}) + (\gamma * \text{Risk_gender})$$

(Where α , β , and γ are customizable weights determining the user's priority).

The Risk Factor Equation:

The safety metric is calculated dynamically based on synthetic incident data and the user's gender profile:

$\text{Risk_gender} = \text{SUM}(\text{Incident_Severity} * \text{Gender_Vulnerability_Multiplier})$

(Example: If an incident like "mugging" is on a road segment, the Gender_Vulnerability_Multiplier might be mathematically higher for a female profile, severely penalizing that path and forcing the AI to search for safer alternative streets).

4. The Main Challenge:

Designing the Heuristic Function, $h(n)$

In standard routing, the heuristic function is simply the Straight-Line Distance (SLD) to the goal. However, because our cost function, $g(n)$, includes a Safety/Risk metric, our heuristic, $h(n)$, must also account for this to remain accurate and efficient.

The primary challenge of this system is designing a heuristic that balances safety while remaining Admissible (it must never overestimate the cost, or A^* will fail to find the optimal path).

Our proposed heuristic equation will combine the Euclidean distance with the absolute minimum possible risk across the map:

$$h(n) = (\alpha * SLD(n, goal)) + (\gamma * Absolute_Minimum_Risk_Remaining)$$

This ensures the AI is guided toward the goal efficiently while respecting the safety parameters without breaking admissibility rules.

5. Methodology and System Pipeline

The project will follow a strict computational pipeline:

1. Map Initialization: Load OSM data and overlay the synthetic incident/risk data points.

2. User Input: The system accepts the Start Node, Goal Node, User Gender, and Weight Preferences (α , β , γ).

3. Execution of Search Algorithms:

- * Uninformed Search: Run an algorithm like Uniform Cost Search (which explores purely based on $g(n)$ blindly).

- * Informed Search: Run A* Search utilizing the newly formulated $h(n)$ heuristic.

4. Result Generation: Output the optimal path onto the map interface.

6. Expected Results and Evaluation Metrics

The success of the system will be evaluated by comparing the outputs of the different search algorithms. The final results will document:

- * Node Exploration Metrics: A quantitative comparison showing how many total nodes were expanded by Uninformed Search versus Informed Search. It is expected that A* will significantly reduce the number of explored nodes.

- * Impact of Heuristic Values (Up-Down Variance): The system will simulate how changing the heuristic weights alters the AI's behavior. For example, if the safety weight (γ) is turned "up," the AI should mathematically justify exploring a longer, more winding path to avoid high-risk nodes.

- * Comparative Trade-offs: An analysis of the trade-off between total path length and total path safety, proving that the system can successfully protect a user at the expense of a marginally longer travel time.